

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/216361330>

Internet Based Information System for ODI Cricket

Conference Paper · October 1998

CITATION

1

READS

12,175

2 authors:



[Ponnampemurage Fernando](#)
Naval Postgraduate School

8 PUBLICATIONS 53 CITATIONS

SEE PROFILE



[Gihan Wikramanayake](#)
University of Colombo

163 PUBLICATIONS 702 CITATIONS

SEE PROFILE

Internet Based Information System for ODI Cricket

P.N.P. Fernando and G.N. Wikramanayake

Department of Statistics and Computer Science
University of Colombo.

E-mail: pnpf@mail.cmb.ac.lk and gihanw@cmb.ac.lk

ABSTRACT

Cricket is a very popular sport in Sri Lanka as well as in other cricket playing countries. Information about a cricket match is provided using a scorecard. The analysis of such scorecards generate information important for players, team management, selection boards and other interested parties since they will lead to better judgement about players, their strengths and weaknesses, and performance as a team. This information is in the form of statistical data that changes with every cricket match.

The World Wide Web is the most commonly used and possibly the largest information system available at present. An Internet based information system using a database as its back-end can effectively manage the cricket data and dynamically present up-to-date statistical information to all its users. The design and development of such an information system for One-Day International (ODI) cricket is described in this paper.

Input of the system is mainly in the form of cricket scorecards. Information processing involve data management, retrieval, producing static or dynamic web pages and answering user requests. Most of the processing is carried out using the database query language. Presentation of statistical information for series, country, match, team statistics, batting records, bowling records, fielding, wicket keeping, player profile and all round performance to the users, and the capability to answer frequently asked questions about cricket are the output of the system.

1. INTRODUCTION

The World Wide Web [BAR95, KRO95] is the most commonly used and possibly the largest information system available at present. There are many web sites available in the Internet for different areas of interest. Most of these sites use static web pages to present information with hyperlinks as creation of such pages are as easy as producing word-processed documents [HES95, MAT98, TUR95].

There are several disadvantages in this procedure such as the inability to cater for specific user needs (e.g. the producing of available information in different user preferred formats) and to update existing pages ensuring of consistency of information. To overcome this the alternate is to use the dynamic web page construction facilities [GUN96, FRA98]. An Internet based information system with a database as the back-end for information storage provides a new dimension, as maintaining consistency of the information and the production of web pages can be done through the database. Such a system should have a central database as the back-end for information storage and a suitable user interface as the front-end for information browsing and querying by external users. Data capturing and maintaining facilities should be provided only to the authorised web master, while the system provides online static and dynamic web page construction facilities, and delayed response retrieval facilities [FER98a]. The suitability of this approach for One-Day International (ODI) cricket and medical consultation data was shown in that paper. This paper describes the design and implementation of an Internet based information system for ODI cricket data using that approach.

1.1 Cricket Data

At international level there are two types of cricket matches played between countries, namely: One-Day International (ODI) and Test cricket. ODI matches are played for a quota of 50 overs per country and they are either played in daytime or day and night, while Test matches are played over 5 days.

ODI matches are more popular than Test matches since there is excitement and the match is over in the same day. ODI cricket is also known as instant cricket since there is always a match result within a period of seven to eight hours unless weather prevents, where as in Test cricket the situation is completely different. This paper is based on the ODI version of this game.

A web site called CricInfo [CRIC] is available in the Internet for cricket. This is a popular web site among cricket loving people since live coverage is given for all the ODI and Test matches with ball by ball description. This site also provides all forms of important information on cricket and contains the archives of all previous cricket matches.

1.2 Cricket Scorecard

Information of a particular cricket match is recorded in a scorecard. In the one-day game the innings of the two teams are recorded. Each innings consist of individual scores of players, how they scored the runs, how they were dismissed, extra runs scored (extras), total runs scored (total), when each batsman was dismissed (fall of wickets) and how the opponent players bowled (bowling figures). The general match information of the type of where, when and how the game was played is also recorded in the scorecard. Basically, a cricket scorecard should be able to describe in detail how a particular game was played. A summarised version of this scorecard is published to describe each match played [FIN85]. These scorecards provide static data once the game is completed, while there are dynamic during the playing time of the match.

Cricket scorecard data are summarised by players, teams, venues, series etc. to produce different forms of cricket statistics and records, such as batting and bowling averages, tournament information, user profiles, highest scores, partnership records and many more. These statistics and records change after each game, and are ideal to demonstrate the management dynamic data on the Internet.

2. DESIGN AND IMPLEMENTATION

All the relevant information in a scorecard of an ODI cricket match should be stored in corresponding database tables in the central database. The database tables of table 1 identify most of them.

Each of these database tables will record detail information. For instance, the table for batting should record the basic (cf. table 2) and additional (cf. table 3) information on batting.

Table Name	General Description of the Contents
Batting	Batsman name, score and other information
Bowling	Bowler name and related information
Fallofwkt	Information of fallen wickets
Match	Match information
Scores	Match scores
Umpire	Information of match umpires
Debut	Information of debut players
Series	Series/Tournament information
Comment	Comments and match related information

Table 1: Database Tables of the Central Cricket Database

Identifier	Description of the Contents
Odino	One Day International Number
Country	Country of the Batsman
Batpos	Batting position of Batsman
Batsman	Name of the Batsman
Fielder	Name of the Fielder Involved in the Dismissal of Batsman
Bowler	Name of the Bowler Who Dismissed The Batsman
Runs	Number of Runs Scored
Howout	How Batsman Got Out, Not Out or Did Not Bat

Table 2: Basic Information on Batting

2.1 Different Styles of Scorecards

Cricket scorecards appear in different styles, depending on the type of information available on it and the method of presenting it to the public. Usually the basic information (e.g. table 2) appears in all scorecards with variations in the format of data. For example, the CricInfo management has been using different formats from time to time and other sites also have used their own data formats. It

is therefore necessary to have an understanding about the structure and contents of the data of these scorecards, if we are to capture the data from these sites.

Identifier	Description of Contents
Balls	Number of Balls faced
Dotballs	Number of Dot balls
Fours	Number of Fours
Sixes	Number of Sixes
Minutes	Number of Minutes played
Ones	Number of Singles
Twos	Number of Twos
Threes	Number of Threes
Caporwkt	Role of the Batsman

Table 3: Additional Information on Batting

2.2 Data Capturing and Processing

Input of the system is mainly in the form of cricket scorecards. Information processing involve data management, retrieval, producing static or dynamic web pages and answering user requests. Most of the processing is carried out using the database query language [CRU95, EME89]. Presentation of statistical information for series, country, match, team statistics, batting records, bowling records, fielding, wicket keeping, player profile and all round performance to the users, and the capability to answer frequently asked questions about cricket are the output of the system.

An automated data capturing process will easy the database maintenance task. Hence a data extraction process from an external site for an ODI cricket scorecard is considered. This process involves capture of important and useful information available in a scorecard. The objective is to process extracted data from a scorecard and automatically update the central database.

Importing such data from a web site into a database is not a direct process. There are several stages involved in this whole exercise. The initial step is to down load the information file from the web site. As the second stage, intermediate files are created for each extracted data set. The data that are to be imported into the central database are taken from these intermediate files.

Such a process can be performed by a text processing language like Perl [WAL96]. The process of preparing these intermediate files using computer programs written in Perl language is described next.

2.2.1 Data Capturing using Perl

Data available in external sites can be either in text or HTML format. Also they can be in different formats and levels. Perl programs should be able to deal with all these types. ODI scorecard for a particular match is the only input necessary for this Perl program and it generates the required output files for various type of information such as batting, bowling, fall of wickets, scores, match, umpire, debut, series and comment. The main objective of this text processing is to extract the necessary available information from the input file (scorecard) to produce different output files for a particular match. The complete process can be operated using a specially developed graphical interface (e.g. a simple visual basic program [GUR95]).

The Perl programs used to extract all the important information from the ODI scorecard will look for keywords available in the scorecard. These keywords play a vital role in order to capture the information available at different locations in the scorecard. This process is illustrated in figure 1 using part of a scorecard. Here, names of the data to be captured are shown in angular brackets and the keywords are highlighted in bold letters.

<i><country></i> Innings <i><batsman></i> <i><fielder></i> <i><bowler></i> <i><runs></i> <i><balls></i> <i><fours></i> <i><sixes></i> <i><minutes></i> <i><ones></i> <i><twos></i> <i><threes></i> <i><dotballs></i> DNB : <i><name1></i> <i><name2></i> <i><name3></i> ... <i><name9></i>
--

Figure 1: Syntax for Batting Information of a Scorecard

It is important to note that the information for the data of tables 2 and 3 are captured through this process. For easy identification of respective matches, the

match identification code and appropriate primary key are noted for each captured data.

Data stored in these tables are used to produce the various results to the Internet users of this information system. It is also important to record information that does not appear in scorecards (e.g. full tournament and team details) as such data are also required for various studies.

Sometimes scorecards can be incomplete. The text processing programs should be able to cope with such situations. An error report should be produced at the end of each data capturing process. Null values or special identification symbols are used when data is not available or missing.

2.2.2 Information Processing

Information available in the database has to be processed in order to produce the results or reports for user requests. Production of these reports involves data retrieval and processing which takes processing time. As a result response to individual request can be delayed since the system has to cater to hundreds or thousands of users at a time. The results that had been requested before can be made available for subsequent users. Even those that can change with time (e.g. career statistics of a player) can be provided immediately as a previous version and subsequently replaced by the latest version. This will prevent users waiting for a response until their information request is processed. Automated refresh of the web page can be done when the processed data is available [FER98b].

3. FINDINGS AND RESULTS

The users of this system can view the information through an interface of a web page. This information is classified into the different categories as in figure 2. We use the batting statistics of this list to highlight the type of details available under a particular category. Such a list is presented in figure 3. The output for

each type of statistics/record is produced as dynamic/static web page as described in section 2.

4. CURRENT STATUS

The system was completely implemented in the Windows NT environment with NT server and Microsoft Access database [BAL97]. The information system was implemented and tested only for ODI data. The statistical information produced by this system can be useful only when the database is updated with all ODI scorecards. Hence, this system is currently not accessible to the public through the Internet. However similar information maintained using our previous systems [SLCP94, SLCP97] is available on the Internet.

Series/Tournament Information (e.g. World Cup) Country Information - Summary results Match Information - Scorecard Team Statistics - Highest/Lowest Totals Batting Statistics - Highest Individual, Partnerships, Most runs Bowling Records - Best Analysis, Hat-tricks, Most wickets Fielding - Most Catches, Innings, Career Wicket Keeping - Most dismissals, Innings, Career Player Profile - Batting, Bowling and Fielding All Round - 1000 Runs and 100 wickets career or 100 dismissals Miscellaneous - Most Appearances, Captain, Country, Matches Ground Profile - Name, Country, records and Match history Umpire Profile - Name, Country and Number of matches

Figure 2: Different Categories of Cricket Statistics

Most ODI Runs in Career Highest ODI Career Batting Averages Highest Individual Scores in ODI's Individual Centuries in ODI Individual Half-Centuries in ODI ODI Partnership Records
--

Figure 3: Different Categories of Batting Statistics

Dynamic web pages of this system are generated using CGI scripts [GUN96]. It is also possible to use Java [JEP97] or active server pages [FRA98] to perform these tasks. It is possible to extend this system to cope with Test

cricket and other form of cricket data (e.g. First Class, School and Club) as in the previous systems. Relevant changes and modifications necessary for this purpose has being carried out [EES98].

5. CONCLUSIONS

Production of dynamic/static web pages with statistical information for series/tournament, country, match, team, batting, bowling, fall of wickets and other related categories of interest is relatively easy when all information of ODI cricket data is available in a central database.

Internet users have the opportunity to get a lot of valuable information regarding ODI cricket from the developed system. This information is useful and important not only to players, but also to the team management for their decisions to select team members.

The information system is available to most users since it is implemented in the Internet environment. People who are interested in cricket will find this information system a useful and informative one.

Once the database is fully updated this system can be also used for various other purposes such as to produce various forms of cricket books, magazines and souvenirs.

6. REFERENCES

- [BAL97] Balter A., "*Mastering Access 97*", Sams Publishing, 1997.
- [BAR95] Barron B., Ellsworth J.H. and Saretz K.M., "*Internet*", Sams. Net Publishing, 1995.
- [CRIC] Cricinfo, "*The home of Cricket*", <http://www.cricket.org>.
- [CRU95] Cruber S., "*Understanding SQL*", BPB Publications, 1995.
- [EME89] Emerson S.L., Darnovsky M. and Bowman J.S., "*The Practical SQL*", Addison-Wesley, 1989.

- [EES98] Eeswara M.P., "*Internet based Statistical Information System for Cricket*", B.Sc. dissertation, University of Colombo, Colombo, Sri Lanka, 1998.
- [FER98a] Fernando P.N.P. and Wikramanayake G.N., "*Designing a Web based Information System with Continuous Growth*", The 17th National Information Technology Conference, Colombo, Sri Lanka, 1998.
- [FER98b] Fernando P.N.P., "*Internet Based Information System for ODI Cricket*", M.Sc. dissertation, University of Colombo, Colombo, Sri Lanka, 1998.
- [FIN85] Findall B. and Isaacs V.H., "The Wisden book of One-Day International Cricket 1971-1985", John Wisden & Co., 1985.
- [FRA98] Francis B., Fedorov A., Harrison R., Sussman D., Homer A., Murphy S. and Smith R., "*Professional Active Server Pages 2.0*", 2nd edition, Wrox Press Inc., 1998.
- [GUN96] Gundavaram S., "*CGI Programming*", O'Reilly & Associates, 1996.
- [GUR95] Gurewicz N. and Gurewicz O., "*Master Visual Basic*", Sams Publishing, 1995.
- [HES95] Heslop B. and Budnick L., "*HTML Publishing on the Internet*", Ventana, 1995.
- [JEP97] Jepson B., "*Java Database Programming*", John Wiley & Sons, 1997.
- [KRO95] Krol E.D. and Ferguson P., "*The Whole Internet*", O'Reilly & Associates, 1995.
- [MAT98] Matthews M.S. and Poulsen E.B., "*Frontpage 98: The Complete Reference*", Osborne McGraw-Hill, 1998.
- [SLCP94] "Sri Lanka Cricket Page", Web site maintained by G.N. Wikramanayake at http://www.cs.cf.ac.uk/Sri_Lanka/Cricket/, 1994-97.
- [SLCP97] "Sri Lanka Cricket Page", Web site maintained by G.N. Wikramanayake at <http://www.cmb.ac.lk/~gihanw/Cricket/>, 1997-98.
- [TUR95] Turligton S.R., "*Walking the World Wide Web*", Ventana, 1995.
- [WAL96] Wall L., Chriansen T. and Schwartz R.L., "*Programming Perl*", O'Reilly, 1996.